

1. NAME OF THE MEDICINAL PRODUCT

Pantoprazole Sandoz Gastro-resistant Tablet 40 mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each gastro-resistant tablet contains 40 mg pantoprazole (as pantoprazole sodium sesquihydrate).

Excipient:

Each gastro-resistant tablet contains 2 µg of the colouring agent Ponceau 4R aluminium lake (E124).

3. PHARMACEUTICAL FORM

Gastro-resistant tablet.

A yellow, oval coated tablet.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

- In combination with two appropriate antibiotics (see "Posology") for the eradication of *Helicobacter pylori* in patients with peptic ulcers with the objective of reducing the recurrence of duodenal and gastric ulcers caused by this microorganism
- Duodenal ulcer
- Gastric ulcer
- Moderate and severe cases of inflammation of the esophagus (reflux esophagitis)
- Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions

4.2 Posology and Method of Administration

Method of administration

Oral use.

Tablets should not be chewed or crushed, and should be swallowed whole with water.

In combination therapy for eradication of *Helicobacter pylori* infection the second tablet should be taken before the evening meal.

Posology

The following information applies unless Pantoprazole 40 mg has been otherwise prescribed by your doctor. Please follow these instructions, as otherwise Pantoprazole 40 mg may not have the desired effect.

Adults and adolescents 12 years of age and above

Treatment of moderate and severe reflux oesophagitis

One tablet of Pantoprazole 40 mg per day. In individual cases, the dose may be doubled (increase to 2 tablets of Pantoprazole 40 mg daily) especially when there has been no response to other treatment.

Adults

Eradication of *H. pylori* in Combination with Two Appropriate Antibiotics

In cases of duodenal or gastric ulcer in which infection with *Helicobacter pylori* has been confirmed, the microorganism should be eradicated by combination treatment.

Depending on the resistance pattern, the following combinations are recommended:

- a) 2 x 1 pantoprazole 40 mg gastro-resistant tablet / day
+ 2 x 1000 mg amoxicillin / day
+ 2 x 500 mg clarithromycin / day
- b) 2 x 1 pantoprazole 40 mg gastro-resistant tablet / day
+ 2 x 500 mg metronidazole / day
+ 2 x 500 mg clarithromycin / day
- c) 2 x 1 pantoprazole 40 mg gastro-resistant tablet / day
+ 2 x 1000 mg amoxicillin / day
+ 2 x 500 mg metronidazole / day

Treatment of gastric and duodenal ulcer

One tablet of Pantoprazole 40 mg per day. In individual cases, the dose may be doubled (increase to 2 tablets of Pantoprazole 40 mg daily) especially when there has been no response to other treatment.

Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions

For the long-term management of Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions patients should start their treatment with a daily dose of 80 mg (2 tablets of Pantoprazole 40 mg). Thereafter, the dose can be titrated up or down as needed using measurements of gastric acid secretion to guide. With doses above 80 mg daily, the dose should be divided and given twice daily. A temporary increase of the dose above 160 mg pantoprazole is possible but should not be applied longer than required for adequate acid control. Treatment duration in Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions is not limited and should be adapted according to clinical needs.

Type and duration of treatment

Combination therapy for eradication of *Helicobacter pylori* infection usually lasts 7 days and can be extended to a maximum of 2 weeks. If after this time further treatment with Pantoprazole 40 mg is indicated to ensure that the ulcer heals completely, the dose recommendations for gastric and duodenal ulcers must be observed.

In the majority of cases, a duodenal ulcer heals completely within 2 weeks. If a two-week treatment period is not sufficient, healing will be achieved in almost all cases within a further 2 weeks.

Gastric ulcers and reflux esophagitis usually require a 4-week course of treatment. If this should be inadequate, healing will in most cases be achieved within a further 4 weeks. Treatment should not exceed 8 weeks as experience with long-term use is limited.

Treatment duration in Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions is not limited and should be adapted according to clinical needs.

Children below 12 years of age

Pantoprazole 40 mg is not recommended for use in children below 12 years of age due to limited data in this age group.

Special Patient Populations

Patients with hepatic impairment

A daily dose of 20 mg pantoprazole (1 tablet of 20 mg pantoprazole) should not be exceeded in patients with severe liver impairment.

Patients with renal impairment

No dose adjustment is necessary in patients with impaired renal function.

4.3 Contraindications

Hypersensitivity to the active substance, substituted benzimidazoles or to any of the excipients of Pantoprazole Sandoz.

Pantoprazole Sandoz must not be used in combination treatment for eradication of *Helicobacter pylori* in patients with moderate to severe hepatic or renal dysfunction, since currently no data are available on the efficacy and safety of Pantoprazole Sandoz in combination treatment of these patients.

4.4 Special warnings and special precautions for use

Combination therapy

In case of combination therapy, the summaries of product characteristics of the respective drugs should be observed.

Influence on vitamin B12 absorption

In patients with Zollinger-Ellison syndrome and other pathological hypersecretory conditions requiring long-term treatment, pantoprazole, as all acid-blocking medicinal products, may reduce the absorption of vitamin B12 (cyanocobalamin) due to hypo- or achlorhydria. This should be considered in patients with reduced body stores or risk factors for reduced vitamin B12 absorption on long-term therapy or if respective clinical symptoms are observed.

Hepatic impairment

In patients with severe liver impairment the liver enzymes should be monitored regularly during treatment with pantoprazole, particularly on long-term use. In the case of a rise of the liver enzymes the treatment should be discontinued.

Gastric malignancy

Symptomatic response to pantoprazole may mask the symptoms of gastric malignancy and may delay diagnosis. In the presence of any alarm symptom (e.g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis, anaemia or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded. Further investigation is to be considered if symptoms persist despite adequate treatment.

Co-administration with HIV protease inhibitors

Co-administration of pantoprazole is not recommended with HIV protease inhibitors for which absorption is dependent on acidic intragastric pH such as atazanavir, nelfinavir due to significant reduction in their bioavailability.

Regular Surveillance

Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Gastrointestinal infections caused by bacteria

Treatment with pantoprazole may lead to a slightly increased risk of gastrointestinal infections caused by bacteria such as *Salmonella* and *Campylobacter* or *C. difficile*.

Severe Cutaneous Adverse Reactions

Severe cutaneous adverse reactions, including erythema multiforme, Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS), and acute generalized exanthematous pustulosis (AGEP) have been reported in association with the use of PPIs. Discontinue pantoprazole at the first signs or symptoms of severe cutaneous adverse reactions or other signs of hypersensitivity and consider further evaluation.

Subacute cutaneous lupus erythematosus (SCLE)

Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Pantoprazole Sandoz. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia

Severe hypomagnesaemia has been reported in patients treated with PPI like Pantoprazole Sandoz for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPI with digoxin or active substances that may cause hypomagnesaemia (e.g. diuretics), healthcare professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Bone fractures

Proton pump inhibitors, especially if used in high doses and over long durations (more than 1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognized risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10 - 40 %. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhoea

Published observational studies suggest that PPI therapy may be associated with an increased risk of *Clostridium difficile* associated diarrhoea, especially in hospitalized patients. This diagnosis should be considered for diarrhoea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency

Daily treatment with acid-suppressing medications over a long period of time (e.g. longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in literature. This diagnosis should be

considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Methotrexate

Concomitant use with high dose methotrexate may elevate and prolong serum levels of methotrexate and/or its metabolite, possibly leading to methotrexate toxicities.

Interference with laboratory tests

Increased chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on chromogranin A level, Pantoprazole Sandoz treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic Properties). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Use in children

To date, there has been no experience with treatment in children. Pantoprazole Sandoz 40 mg is not recommended for use in children less than 12 years due to the limited experience in this age group.

This medicinal product contains colouring agent Ponceau 4R Aluminium Lake (E 124), which may cause allergic reactions.

4.5 Interaction with other medicinal products and other forms of interaction

Medicinal products with pH-dependent absorption pharmacokinetics

Because of profound and long lasting inhibition of gastric acid secretion, pantoprazole may interfere with the absorption of other medicinal products where gastric pH is an important determinant of oral bioavailability, e.g. some azole antifungals such as ketoconazole, itraconazole, posaconazole and other medicines such as erlotinib.

HIV protease inhibitors

Co-administration of pantoprazole is not recommended with HIV protease inhibitors for which absorption is dependent on acidic intragastric pH such as atazanavir, nelfinavir due to significant reduction in their bioavailability.

If the combination of HIV protease inhibitors with a proton pump inhibitor is judged unavoidable, close clinical monitoring is recommended. Dose of the HIV protease inhibitors may need to be adjusted.

Coumarin anticoagulants (phenprocoumon or warfarin)

Co-administration of pantoprazole with warfarin or phenprocoumon did not affect the pharmacokinetics of warfarin, phenprocoumon or INR. However, there have been reports of increased INR and prothrombin time in patients receiving PPIs and warfarin or phenprocoumon concomitantly. Increases in INR and prothrombin time may lead to abnormal bleeding, and even death. Patients treated with pantoprazole and warfarin or phenprocoumon may need to be monitored for increase in INR and prothrombin time.

Other interactions studies

Pantoprazole is extensively metabolized in the liver via the cytochrome P450 enzyme system.

The main metabolic pathway is demethylation by CYP2C19 and other metabolic pathways include oxidation by CYP3A4.

Interaction studies with active substances also metabolized with these pathways, like carbamazepine, diazepam, glibenclamide, nifedipine, and an oral contraceptive containing levonorgestrel and ethinyl oestradiol did not reveal clinically significant interactions.

An interaction of pantoprazole with other medicinal products or compounds, which are metabolised using the same enzyme system, cannot be excluded.

Results from a range of interaction studies demonstrate that pantoprazole does not affect the metabolism of active substances metabolised by CYP1A2 (such as caffeine, theophylline), CYP2C9 (such as piroxicam, diclofenac, naproxen), CYP2D6 (such as metoprolol), CYP2E1 (such as ethanol) or does not interfere with p-glycoprotein related absorption of digoxin.

There were no interactions with concomitantly administered antacids.

Interaction studies have also been performed by concomitantly administering pantoprazole with the respective

antibiotics (clarithromycin, metronidazole, amoxicillin) No clinically relevant interactions were found.

Medicinal products that inhibit or induce CYP2C19

Concomitant administration of pantoprazole and tacrolimus may increase whole blood levels of tacrolimus, especially in transplant patients who are intermediate or poor metabolizers of CYP2C19.

Inhibitors of CYP2C19 such as fluvoxamine could increase the systemic exposure of pantoprazole. A dose reduction may be considered for patients treated long-term with high doses of pantoprazole, or those with hepatic impairment.

Enzyme inducers affecting CYP2C19 and CYP3A4 such as rifampicin and St John's wort (*Hypericum perforatum*) may reduce the plasma concentrations of PPIs that are metabolised through these enzyme systems.

Methotrexate

Concomitant use of high dose methotrexate (e.g. 300 mg) and proton-pump inhibitors has been reported to increase methotrexate levels in some patients. Therefore in settings where high-dose methotrexate is used, for example cancer and psoriasis, a temporary withdrawal of pantoprazole may need to be considered.

4.6 Pregnancy and lactation

Pregnancy

A moderate amount of data on pregnant women (between 300-1000 pregnancy outcomes) indicate no malformative or foeto/ neonatal toxicity of pantoprazole.

Animal studies have shown reproductive toxicity.

As a precautionary measure, it is preferable to avoid the use of pantoprazole during pregnancy.

Breast-feeding

Animal studies have shown excretion of pantoprazole in breast milk.

There is insufficient information on the excretion of pantoprazole in human milk but excretion into human milk has been reported. A risk to the newborns/infants cannot be excluded.

Therefore, a decision on whether to discontinue breast-feeding or to discontinue/abstain from pantoprazole therapy taking into account the benefit of breast-feeding for the child, and the benefit of pantoprazole therapy for the woman.

Fertility

There was no evidence of impaired fertility following the administration of pantoprazole in animal studies.

4.7 Effects on ability to drive and use machines

Pantoprazole has no or negligible influence on the ability to drive and use machines.

Adverse drug reactions such as dizziness and visual disturbances may occur (see section 4.8).

If affected, patients should not drive or operate machines.

4.8 Undesirable Effects

Summary of the safety profile

Approximately 5% of patients can be expected to experience adverse drug reactions (ADRs). The most commonly reported ADRs are diarrhoea and headache, both occurring in approximately 1% of patients.

The list below shows adverse reactions reported with pantoprazole, ranked under the following frequency classification: Very common; common; uncommon; rare; very rare; not known.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System Organ Class	Frequency	Adverse Events
Blood and lymphatic system disorders	Rare	Agranulocytosis
	Very rare	Thrombocytopenia, leukopenia, pancytopenia
Immune system disorders	Rare	Hypersensitivity (including anaphylactic reactions and anaphylactic shock)
Metabolism and nutrition disorders	Rare	Hyperlipidaemias and lipid increases (triglycerides, cholesterol), weight changes

	Not known	Hyponatraemia, hypomagnesaemia, hypocalcaemia in association with hypomagnesemia, hypokalaemia in association with hypomagnesemia
	-	Vitamin B12 deficiency
Psychiatric disorders	Uncommon	Sleep disorders
	Rare	Depression (and all aggravations)
	Very rare	Disorientation (and all aggravations)
	Not known	Hallucination, confusion (especially in pre-disposed patients, as well as the aggravation of these symptoms in case of pre-existence)
Nervous system disorders	Uncommon	Headache, dizziness
	Rare	Taste disorders
	Not known	Paraesthesia
Eye disorders	Rare	Disturbances in vision/blurred vision
Gastrointestinal disorders	Common	Fundic gland polyps (benign)
	Uncommon	Diarrhoea, nausea/vomiting, abdominal distension and bloating, constipation, dry mouth, abdominal pain and discomfort
	Not known	Microscopic colitis
Hepatobiliary disorders	Uncommon	Liver enzymes increased (transaminases, γ -GT).
	Rare	Bilirubin increased
	Not known	Hepatocellular injury, jaundice, hepatocellular failure
Skin and subcutaneous tissue disorders	Uncommon	Rash/exanthema/eruption, pruritus
	Rare	Urticaria, angioedema
	Not known	Stevens-Johnson syndrome, Lyell syndrome or toxic epidermal necrolysis (TEN), erythema multiforme, photosensitivity, subacute cutaneous lupus erythematosus, drug reaction with eosinophilia and systemic symptoms (DRESS), acute generalized exanthematous pustulosis
Musculoskeletal and connective tissue disorders	Uncommon	Fracture of the hip, wrist or spine
	Rare	Arthralgia, myalgia
	Not known	Muscle spasm as a consequence of electrolyte disturbances
Renal and urinary disorders	Not known	Interstitial nephritis (with possible progression to renal failure)
Reproductive system and breast disorders	Rare	Gynaecomastia
General disorders and administration site conditions	Uncommon	Asthenia, fatigue and malaise
	Rare	Body temperature increased, oedema peripheral
Infections and infestations	-	<i>Clostridium difficile</i> associated diarrhoea

4.9 Overdose

There are no known symptoms of overdose in man.

Systemic exposure with up to 240 mg administered intravenously over 2 minutes was well tolerated.

As pantoprazole is extensively protein bound, it is not readily dialysable.

In the case of overdose with clinical signs of intoxication, apart from symptomatic and supportive treatment, no specific therapeutic recommendations can be made.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Pharmacotherapeutic group: Proton pump inhibitors, ATC Code: A02BC02.

Mechanism of action

Pantoprazole is a substituted benzimidazole which inhibits the secretion of hydrochloric acid in the stomach by specific blockade of the proton pumps of the parietal cells.

Pantoprazole is converted to its active form in the acidic environment in the parietal cells where it inhibits the H⁺, K⁺-ATPase enzyme, i.e. the final stage in the production of hydrochloric acid in the stomach.

The inhibition is dose-dependent and affects both basal and stimulated acid secretion. In most patients, freedom from symptoms is achieved within 2 weeks. As with other proton pump inhibitors and H₂ receptor inhibitors, treatment with pantoprazole reduces acidity in the stomach and thereby increases gastrin in proportion to the reduction in acidity. The increase in gastrin is reversible. Since pantoprazole binds to the enzyme distal to the cell receptor level, it can inhibit hydrochloric acid secretion independently of stimulation by other substances (acetylcholine, histamine, gastrin). The effect is the same whether the product is given orally or intravenously.

The fasting gastrin values increase under pantoprazole. On short-term use, in most cases they do not exceed the upper limit of normal. During long-term treatment, gastrin levels double in most cases. An excessive increase, however, occurs only in isolated cases. As a result, a mild to moderate increase in the number of specific endocrine (ECL) cells in the stomach is observed in a minority of cases during long-term treatment (simple to adenomatoid hyperplasia).

During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.

Available published evidence suggests that proton pump inhibitors (PPI) should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

An influence of a long-term treatment with pantoprazole exceeding one year cannot be completely ruled out on endocrine parameters of the thyroid according to results in animal studies.

5.2 Pharmacokinetic Properties

Absorption

Pantoprazole is rapidly absorbed and the maximal plasma concentration is achieved even after one single 40 mg oral dose. On average at about 2.5 h p.a. the maximum serum concentrations of about 2 - 3 µg/ml are achieved, and these values remain constant after multiple administration.

Pharmacokinetics do not vary after single or repeated administration. In the dose range of 10 to 80 mg, the plasma kinetics of pantoprazole are linear after both oral and intravenous administration.

The absolute bioavailability from the tablet was found to be about 77%. Concomitant intake of food had no influence on AUC, maximum serum concentration and thus bioavailability. Only the variability of the lag-time will be increased by concomitant food intake.

Distribution

Pantoprazole's serum protein binding is about 98%. Volume of distribution is about 0.15 l/kg.

Biotransformation

The substance is almost exclusively metabolized in the liver. The main metabolic pathway is demethylation by CYP2C19 with subsequent sulphate conjugation, other metabolic pathway include oxidation by CYP3A4.

Elimination

Terminal half-life is about 1 hour and clearance is about 0.1 l/h/kg. Because of the specific binding of pantoprazole to the proton pumps of the parietal cell the elimination half-life does not correlate with the much longer duration of action (inhibition of acid secretion).

Renal elimination represents the major route of excretion (about 80%) for the metabolites of pantoprazole, the rest is excreted with the faeces. The main metabolite in both the serum and urine is desmethylpantoprazole which is conjugated with sulphate. The half-life of the main metabolite (about 1.5 hours) is not much longer than that of pantoprazole.

Special populations

Renal Impairment

No dose reduction is recommended when pantoprazole is administered to patients with impaired renal function (including dialysis patients). Only very small amounts of pantoprazole are dialyzed. Although the main metabolite

has a moderately delayed half-life (2 - 3h), excretion is still rapid and thus accumulation does not occur.

Hepatic impairment

Although for patients with liver cirrhosis, (classes A and B according to Child), the half-life values increased to between 7 to 9 h, and the AUC values increased by a factor of 5-7 (pantoprazole 40 mg), the maximum serum concentration only increased slightly by a factor of 1.5 (pantoprazole 40 mg) compared with healthy subjects.

Elderly

A slight increase in AUC and C_{max} in elderly volunteers compared with younger counterparts is also not clinically relevant.

6. PHARMACEUTICAL PARTICULARS

6.1 Incompatibilities

Not applicable.

6.2 Shelf life

2 years.

6.3 Special Precautions for Storage

Do not store above 30°C.

6.4 Dosage forms and packaging available

Pantoprazole Sandoz Gastro-resistant Tablet 40mg - 7 tablets/ box and 14 tablets/ box.

Not all pack sizes may be marketed.

7. PRODUCT REGISTRATION HOLDER

Sandoz Products Malaysia Sdn. Bhd.

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8. DATE OF REVISION OF THE TEXT

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